

MI-07: Resolution

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Repository status: Solved. The complete argument received an independent Codex-agent PASS review against the exact catalog target. The original draft was AI-assisted; the reviewed proof below is unchanged. This is independent agent verification, not external human peer review or formal proof certification. [Detailed review and scope](#).

1 MI-07: no finite two-unitary constant for the maximal symmetric modulus

For a square matrix Z , define its maximal symmetric modulus by

$$\mathcal{M}(Z) = \lim_{r \rightarrow \infty} (|Z|^r + |Z^*|^r)^{1/r},$$

where r runs through positive integers. This is the spectral-order join $|Z| \vee |Z^*|$. The target is the constant-one assertion of [1, Conjecture 3.12].

Theorem 1.1. *There is no finite nonnegative constant C , even in dimension two, such that every $A, B \in \mathbb{M}_2(\mathbb{C})$ admits unitaries U, V with*

$$\mathcal{M}(A + B) \leq C(U\mathcal{M}(A)U^* + V\mathcal{M}(B)V^*). \quad (1)$$

In particular, the proposed constant $C = 1$ is false.

Proof. Let $t > 0$ and take

$$A = e_1 e_1^* = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad B = t e_1 e_2^* = \begin{pmatrix} 0 & t \\ 0 & 0 \end{pmatrix}.$$

Since A is a projection,

$$(|A|^r + |A^*|^r)^{1/r} = 2^{1/r} A, \quad \mathcal{M}(A) = A.$$

For B , the right and left moduli are $t e_2 e_2^*$ and $t e_1 e_1^*$, so

$$\mathcal{M}(B) = t I_2.$$

Put $s = \sqrt{1 + t^2}$ and $v = (e_1 + t e_2)/s$. The rank-one matrix $A + B = e_1(e_1 + t e_2)^*$ has moduli

$$|A + B| = s v v^*, \quad |(A + B)^*| = s e_1 e_1^*.$$

The matrix $R = v v^* + e_1 e_1^*$ is positive definite because v and e_1 are linearly independent. Consequently

$$(|A + B|^r + |(A + B)^*|^r)^{1/r} = s R^{1/r} \longrightarrow s I_2,$$

and therefore $\mathcal{M}(A + B) = s I_2$.

For arbitrary U, V , the matrix

$$U\mathcal{M}(A)U^* + V\mathcal{M}(B)V^* = Ue_1e_1^*U^* + tI_2$$

has eigenvalues $1 + t$ and t . Taking a unit vector orthogonal to Ue_1 in (1) forces

$$s \leq Ct, \quad C \geq \frac{\sqrt{1+t^2}}{t}.$$

This lower bound is unbounded as $t \downarrow 0$. For $C = 1$, it is already impossible for every $t > 0$. $\square$

Remark 1.2. One may choose $t = 1/2$ for a fixed explicit counterexample: $\mathcal{M}(A+B) = (\sqrt{5}/2)I_2$, whereas every two-orbit sum on the right before multiplication by C has eigenvalues $3/2$ and $1/2$. The proof also answers negatively the dimension-dependent finite-constant question preceding the conjecture [1, Question 3.11]. The calculation uses the spectral-order join, not the usual pointwise maximum of entries or a different symmetric modulus.

References

- [1] Jean-Christophe Bourin and Eun-Young Lee. Averages over matrix unitary orbits and spectral order. arXiv:2606.15624, 2026. Version consulted: v2; Question 3.11, Conjecture 3.12, and Questions 4.8–4.9. URL: <https://arxiv.org/abs/2606.15624>.